Immune Dysregulation With Immunodeficiency Due To AIOLOS Haploinsufficiency (MONDO:0979328)

Causal gene: *IKZF3* (encoding the Ikaros-family zinc-finger transcription factor AIOLOS)

Summary

Immune Dysregulation With Immunodeficiency Due To AIOLOS Haploinsufficiency (MONDO:0979328) is a rare, autosomal-dominant **inborn error of immunity (IEI)** caused by heterozygous germline variants in ***IKZF3***, the gene encoding the lymphoid-restricted Ikaros-family zinc-finger transcription factor **AIOLOS**. Despite the disease name emphasizing "haploinsufficiency," the broader *IKZF3*-associated disease encompasses two mechanistically distinct classes of variant: (1) **DNA-binding-domain dominant-negative variants** (G159R, N160S, G191R) that abolish DNA binding and pericentromeric heterochromatin targeting and poison wild-type AIOLOS, and (2) **true haploinsufficiency variants** (E82K, Q402*) that reduce functional AIOLOS dosage via protein instability or truncation. Both converge on defective lymphocyte transcriptional programs and produce a combined immunodeficiency with immune dysregulation.

Clinically, affected individuals present with **recurrent sinopulmonary and opportunistic infections** (including *Pneumocystis jirovecii* pneumonia), **hypogammaglobulinemia with abnormal B- and T-cell development**, **autoimmunity and allergy**, and an elevated risk of **B-cell malignancy** (chronic lymphocytic leukemia, CLL). Severity and penetrance are variant-dependent: the original DNA-binding dominant-negative N160S variant was fully penetrant and segregated through a multigeneration family, whereas the haploinsufficiency variants (E82K, Q402*) show incomplete clinical penetrance*. A fatal case of disseminated vaccine-derived rubella virus (VDRV) granulomatous disease underscores that live attenuated vaccines are contraindicated and that prognosis can be grave.

The disease sits within the broader **Ikaros zinc-finger family (IKZF1/Ikaros, IKZF2/Helios, IKZF3/Aiolos)** spectrum of IEIs that share molecular readouts — loss of DNA binding, loss of pericentromeric targeting, and disrupted dimerization/NuRD-complex function. The human phenotype has been **recapitulated in *Ikzf3* mouse models**, including an Aiolos-null model that develops SLE-like autoimmunity and a humanized knock-in carrying the patient N160S-equivalent mutation. Management follows general combined-immunodeficiency standards (immunoglobulin replacement, antimicrobial prophylaxis, immunomodulation, and allogeneic HSCT for severe phenotypes), with malignancy surveillance, genetic counseling, and strict avoidance of live vaccines.

Key Findings

Finding 1 — The disease is caused by heterozygous *IKZF3* (AIOLOS) variants

AIOLOS-associated immunodeficiency (MONDO:0979328) is caused by **germline heterozygous variants in *IKZF3***, which encodes the Ikaros-family C2H2 zinc-finger transcription factor **AIOLOS**. The original kindred, described by Kuehn et al. (2021), carried the missense variant **AIOLOS p.N160S** and presented as a novel combined immunodeficiency with T- and B-cell involvement, *Pneumocystis jirovecii* pneumonia, and/or CLL. The N160S mutant **failed DNA binding and pericentromeric targeting** and acted as a **dominant negative over wild-type AIOLOS but not wild-type IKAROS**; the variant was **fully penetrant** in that family.

> "Here we describe a novel combined immunodeficiency due to an IKZF3 mutation in a family presenting with T and B cell involvement, Pneumocystis jirovecii pneumonia, and/or chronic lymphocytic leukemia." — PMID: 34694366
>
> "The mutant was fully penetrant and had a dominant-negative effect over WT AIOLOS but not WT IKAROS." — PMID: 34694366

Evidence source: human clinical + in vitro functional studies.

Finding 2 — Two mechanistic classes of AIOLOS variants: DNA-binding dominant-negative vs. haploinsufficiency

A review by Yamashita & Morio (2024) and the mechanistic study by Kuehn et al. (2024) establish **two distinct variant classes**:

Variant class	Variants	Protein domain affected	Mechanism	Penetrance
DNA-binding loss-of-function (dominant-negative)	G159R, N160S, G191R	DNA-binding zinc fingers (ZF2–3)	Abolish DNA binding and pericentromeric targeting; poison WT AIOLOS	Full (N160S)
Haploinsufficiency	E82K (missense), Q402* (truncating)	Protein-stability domain (E82K); dimerization ZF5–6 lost (Q402*)	Reduced AIOLOS dosage via instability or truncation	Incomplete

Notably, Kuehn et al. (2024) reported **9 individuals from 3 unrelated families** carrying Q402 *or E82K that cause haploinsufficiency through different mechanisms. The Q402* truncation lacks the ZF5–6 dimerization domain yet still homodimerizes with WT AIOLOS and **negatively regulates DNA binding through ZF1** — a newly recognized AIOLOS function. E82K affects a **redefined protein-stability domain**.

> "heterozygous missense loss-of-function variants within the DNA-binding domain of the IKZF3 gene (G159R, N160S, and G191R) have been identified in patients with inborn errors of immunity (IEI). Additionally, a missense and a truncating variant (E82K and Q402X) leading to the AIOLOS haploinsufficiency have been documented." — PMID: 38773004
>
> "we report on 9 individuals from 3 unrelated families carrying AIOLOS variants Q402* or E82K, which led to AIOLOS haploinsufficiency through different mechanisms of action." — PMID: 38015619
>
> "Patients with AIOLOS haploinsufficiency showed hypogammaglobulinemia, recurrent infections, autoimmunity, and allergy, but with incomplete clinical penetrance." — PMID: 38015619

Evidence source: human clinical + in vitro functional studies.

Finding 3 — *Ikzf3* mouse models recapitulate immunodeficiency and SLE-like autoimmunity

Aiolos-null mice develop **systemic lupus erythematosus (SLE)-like disease** with anti-dsDNA antibodies, immune-complex–mediated glomerulonephritis, B-cell hyperproliferation, elevated B-cell activation markers, and spontaneous germinal centers (Sun et al., 2003). A **knock-in murine model carrying the human N160S-equivalent mutation recapitulated the human immunophenotype** (Kuehn et al., 2021), validating causality.

> "mice lacking the zinc finger transcription factor Aiolos develop the symptoms of human systemic lupus erythematosus (SLE), which is characterized by the production of anti-dsDNA Ab and immune complex-mediated glomerulonephritis." — PMID: 12574333
>
> "The human immunophenotype was recapitulated in a murine model carrying the corresponding human mutation." — PMID: 34694366

Evidence source: model organism (mouse).

Finding 4 — Clinical phenotype: recurrent infections, immune dysregulation, and malignancy risk

The majority of individuals with AIOLOS-associated IEI manifest **recurrent sinopulmonary infections** plus various bacterial and viral infections. Key documented features across cases:

- **Opportunistic infection** (*Pneumocystis jirovecii* pneumonia) and **progression to CLL** in the index N160S family (Kuehn 2021).
- A **de novo G191R** pediatric patient (age 1 y 4 mo) with recurrent infections, diarrhea, failure to thrive, decreased T lymphocytes and immunoglobulins, and reduced recent thymic emigrants among naive CD4 T cells (Shi 2024).
- **Haploinsufficiency** (E82K, Q402*) patients with hypogammaglobulinemia, recurrent infections, autoimmunity, and allergy (incomplete penetrance; Kuehn 2024).
- A **fatal disseminated vaccine-derived rubella virus (VDRV)** granulomatous disease with multi-organ failure and CNS involvement in an *IKZF3*-mutant patient (Zhou 2026).

> "The majority of individuals with AIOLOS-associated IEI manifest recurrent sinopulmonary infections, as well as various bacterial and viral infections." — PMID: 38773004
>
> "we evaluated a 1-year-4-month-old female patient presenting with recurrent infections, diarrhea, and failure to thrive. Laboratory investigations revealed decreased T lymphocyte and immunoglobulin levels." — PMID: 38758229
>
> "a case of disseminated granulomatous disease associated with live attenuated vaccine-derived rubella virus (VDRV) in an immunodeficient patient who subsequently developed fatal multi-organ failure, including neurological deficits, with VDRV detected in the cerebrospinal fluid" — PMID: 41746515

Evidence source: human clinical.

Finding 5 — AIOLOS disease sits within the Ikaros-family (IKZF1/2/3) spectrum of IEIs

Germline mutations across the Ikaros zinc-finger family cause phenotypically overlapping IEIs:

- **IKZF1 (Ikaros)** haploinsufficiency causes autosomal-dominant CVID (e.g., p.His167Tyr; confirmed deleterious via impaired pericentromeric heterochromatin localization and EMSA — a haploinsufficiency mechanism; Stojkic 2024).
- **IKZF2 (Helios)** monoallelic mutations cause immune dysregulation (SLE, ITP, EBV-HLH) with hypogammaglobulinemia and decreased Tfh/NK cells; mutants cannot homodimerize/bind DNA and show aberrant NuRD-complex interactions (Shahin 2022).
- **IKZF3 (AIOLOS)** causes the present disease.

The **shared molecular readouts** are loss of DNA binding, loss of pericentromeric (heterochromatin) targeting, and disrupted dimerization/NuRD-complex function.

> "whole genome sequencing revealed a novel, pathogenic c.499C > T: p.His167Tyr variant in IKZF1 ... Functional testing done through pericentromeric heterochromatin localization and light shift chemiluminescent electrophoretic mobility shift assay confirmed the variant's deleterious effect via a haploinsufficiency mechanism." — PMID: 38734037

> "germline mutations in IKZF1 and IKZF3 encoding Ikaros and Aiolos have recently been identified in patients with phenotypically similar immunodeficiency syndromes" — PMID: 34920454
>
> "Functional assays revealed the inability of HeliosR291X to homodimerize and bind target DNA as dimers." — PMID: 34920454

Evidence source: human clinical + in vitro.

Finding 6 — AIOLOS molecular function and the mechanistic causal chain

AIOLOS/IKZF3 is a **C2H2 zinc-finger transcription factor highly expressed in the lymphoid lineage**, critical for lymphocyte differentiation and development. Mechanistically it (i) binds DNA via its central zinc fingers, (ii) homodimerizes/heterodimerizes (with IKAROS) via C-terminal zinc fingers, (iii) targets pericentromeric heterochromatin, and (iv) operates with the **NuRD chromatin-remodeling complex**. Disease variants in the DNA-binding domain (N160S, G159R, G191R) abolish DNA binding and pericentromeric targeting and act dominant-negatively over WT AIOLOS; haploinsufficiency variants (E82K, Q402*) reduce dosage. Downstream, AIOLOS positively regulates CXCR3 on Th1 cells via an IFN-γ/STAT1 feedback loop (Leonard 2024), restrains intestinal intraepithelial lymphocyte activation (Yomogida 2024), and shapes human Th17 functional states** together with Eos/IKZF (Cerboni 2026).

> "In vitro studies demonstrated that the mutant protein failed DNA binding and pericentromeric targeting." — PMID: 34694366
>
> "Patients carrying the AIOLOS p.N160S heterozygous variant displayed impaired humoral responses, abnormal B cell development (high percentage of CD21low B cells and negative CD23 expression), and abrogated CD40 responses." — PMID: 34694366
>
> "we identify the Ikaros zinc finger (IkZF) transcription factor Aiolos (Ikzf3) as an additional positive regulator of CXCR3 both in vitro and in vivo using a murine model of influenza virus infection." — PMID: 39560988

Evidence source: human clinical + in vitro + model organism.

Finding 7 — Autosomal dominant inheritance, variable penetrance; ultra-rare, both familial and de novo

AIOLOS-associated disease is **autosomal dominant**, caused by heterozygous (monoallelic) *IKZF3* germline variants. Both inherited and de novo origins are documented:

- **N160S** segregated through a multigeneration family and was fully penetrant (Kuehn 2021).
- **E82K and Q402*** were reported in 9 individuals across 3 unrelated families with incomplete penetrance (Kuehn 2024).
- **G191R** arose **de novo** in a sporadic pediatric case (Shi 2024).

The disease is **ultra-rare**: as of 2024–2026, only a few families and **fewer than ~30 affected individuals** carrying a handful of recurrent DNA-binding-domain (G159R, N160S, G191R) and haploinsufficiency (E82K, Q402*) variants have been reported worldwide.

> "we report on 9 individuals from 3 unrelated families carrying AIOLOS variants Q402* or E82K" — PMID: 38015619
>
> "we discovered a de novo mutation in IKZF3 (NM_012481; exon 5 c.571G > C, p.Gly191Arg)" — PMID: 38758229

Evidence source: human clinical.

Finding 8 — Management, prognosis, and prevention

No disease-specific therapeutic trials exist; **management follows general combined-immunodeficiency/IEI standards**: immunoglobulin replacement (IVIG/SCIG) for hypogammaglobulinemia, antimicrobial prophylaxis (including anti-*Pneumocystis* trimethoprim-sulfamethoxazole given documented PCP), immunomodulation for autoimmune manifestations, and allogeneic HSCT for severe combined-immunodeficiency phenotypes (potentially curative; inferred from IEI practice). Prognosis is **variable and can be fatal** — disseminated VDRV granulomatous disease produced fatal multi-organ failure with CNS involvement, and the index family developed CLL. Prevention hinges on **genetic counseling** (autosomal dominant, 50% transmission risk, variable/incomplete penetrance, de novo possible), cascade testing, malignancy surveillance, and **strict avoidance of live attenuated vaccines**.

A mechanistic caveat for therapy: the immunomodulatory drugs (IMiDs) / CELMoDs (lenalidomide, pomalidomide, iberdomide, mezigdomide) **degrade** AIOLOS via cereblon — the **opposite** of the therapeutic need in this loss-of-function disease — and are therefore mechanistically contraindicated.

> "a family presenting with T and B cell involvement, Pneumocystis jirovecii pneumonia, and/or chronic lymphocytic leukemia" — PMID: 34694366
>
> "an immunodeficient patient who subsequently developed fatal multi-organ failure, including neurological deficits, with VDRV detected in the cerebrospinal fluid" — PMID: 41746515
>
> "two critical lymphoid transcription factors, Ikaros family zinc finger proteins 1 and 3 (IKZF1 and IKZF3), also known as Ikaros and Aiolos" — PMID: 38539501

Evidence source: human clinical + inferred from IEI practice.

Report by Template Section

1. Disease Information

Overview. AIOLOS-associated immunodeficiency / immune dysregulation is a Mendelian, autosomal-dominant inborn error of immunity caused by heterozygous germline variants in ***IKZF3*** (AIOLOS). It is a **combined immunodeficiency with immune dysregulation**: defective humoral and cellular immunity producing recurrent/opportunistic infections, hypogammaglobulinemia, autoimmunity, allergy, and predisposition to B-cell malignancy.

Key identifiers. - **MONDO:** MONDO:0979328 - **Gene:** *IKZF3* (HGNC:13178; NCBI Gene 22806; chromosome 17q12-q21) - **Protein:** AIOLOS (UniProt Q9UKT9) - **OMIM/Orphanet/ICD/MeSH:** Not recorded in the provided identifiers; classified broadly under inborn errors of immunity / primary immunodeficiency. (The IUIS IEI classification groups it with the Ikaros-family transcription-factor defects.)

Synonyms / alternative names. "AIOLOS haploinsufficiency," "AIOLOS-associated inborn errors of immunity (IEI)," "IKZF3-related combined immunodeficiency," "Immune dysregulation with immunodeficiency due to AIOLOS haploinsufficiency."

Information source. Derived from **aggregated disease-level resources and primary case reports / case series** (not EHR cohorts). The evidence base is small (a handful of families; <~30 individuals).

2. Etiology

Primary cause: monogenic — heterozygous germline *IKZF3* variants. **Genetic risk = the causal variant itself**; there are no established environmental or infectious causes. **Risk factors** are a positive family history (autosomal dominant) and inheritance of a pathogenic allele; de novo variants also occur. **No validated protective genetic/environmental factors** are documented. **Gene–environment interaction:** the main interaction of clinical consequence is iatrogenic/infectious — exposure to **live attenuated vaccines** (e.g., rubella) in a genetically immunodeficient host leads to disseminated vaccine-derived infection (VDRV;).

P 41746515

Pathogen exposure drives the infectious phenotype in the context of the genetic defect.

3. Phenotypes

Phenotype	Type	Onset	Frequency	Suggested HPO
Recurrent sinopulmonary/ bacterial/viral infections	Clinical sign	Childhood (often)	Majority	HP:0002721 (Recurrent respiratory infections), HP:0002719 (Recurrent infections)
Hypogammaglobulinemia / impaired humoral responses	Lab abnormality	Childhood	Common	HP:0004313 (Decreased Ig level), HP:0002720 (Decreased IgG)
Abnormal B-cell development (↑CD21low, ↓CD23, abrogated CD40)	Lab abnormality	Childhood	N160S family	HP:0010976 (B-cell abnormality)
Decreased T lymphocytes / reduced recent thymic emigrants	Lab abnormality	Infancy (G191R)	Reported	HP:0005403 (Decreased T cells)
Pneumocystis jirovecii pneumonia	Clinical sign	Variable	Index family	Opportunistic infection
Autoimmunity (e.g., cytopenias, SLE-like)	Clinical sign	Variable	Haploinsufficiency class	HP:0002960 (Autoimmunity)
Allergy	Clinical sign	Variable	Haploinsufficiency class	HP:0012393 (Allergy)
Chronic lymphocytic leukemia	Clinical sign	Adult	Index family	HP:0005550 (B-cell lymphoproliferation)
Failure to thrive, diarrhea	Clinical sign	Infancy	G191R case	HP:0001508 (Failure to thrive), HP:0002014 (Diarrhea)

Severity/progression: variable; ranges from mild/incompletely penetrant (E82K, Q402*) to severe/fatal (VDRV case). Quality-of-life impact:** chronic infection burden, immunoglobulin dependence, autoimmune morbidity, and malignancy risk substantially affect daily functioning; formal QOL instruments have not been applied to this ultra-rare disorder.

4. Genetic / Molecular Information

- **Causal gene:** *IKZF3* / AIOLOS (HGNC:13178).
- **Reported pathogenic variants:**
- **DNA-binding domain (dominant-negative):** G159R, **N160S**, G191R (c.571G>C, de novo) — missense.
- **Haploinsufficiency: E82K** (missense, protein-stability domain), **Q402*** (nonsense/truncating, loses ZF5–6 dimerization domain).
- **Classification:** pathogenic / likely pathogenic by functional evidence (ACMG PS3-type functional data: loss of DNA binding, loss of pericentromeric targeting, EMSA).
- **Allele frequency:** essentially absent from population databases (ultra-rare; private/de novo).
- **Origin:** germline (familial segregation and de novo both documented).
- **Functional consequences:** dominant-negative loss-of-function (DNA-binding variants) vs. haploinsufficiency (dosage reduction).
- **Modifier genes:** none validated; the paralogous **IKZF1/IKZF2** define a shared functional network. **Epigenetic/chromosomal abnormalities:** not a feature (point variants, not structural).

5. Environmental Information

No toxic, occupational, lifestyle, or dietary environmental causes are established. The relevant **infectious agents** are opportunistic/vaccine-derived pathogens that exploit the immunodeficiency: *Pneumocystis jirovecii* (PCP) and **vaccine-derived rubella virus (VDRV)**, plus assorted bacterial and viral infections.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **heterozygous germline *IKZF3* variant** arises (inherited or de novo) → produces an abnormal AIOLOS protein.

2. **Branch A (DNA-binding variants: N160S, G159R, G191R):** the mutant **loses DNA binding and pericentromeric heterochromatin targeting** and **dimerizes with wild-type AIOLOS**, *leading to* a **dominant-negative** poisoning of residual WT AIOLOS function (demonstrated in vitro). **Branch B (haploinsufficiency variants: E82K, Q402*):** the variant **reduces functional AIOLOS dosage** — E82K via **protein instability**, Q402 *via truncation (loss of ZF5–6) that nonetheless retains homodimerization and negatively regulates DNA binding through ZF1** (demonstrated in vitro).
3. Reduced/abrogated AIOLOS transcriptional activity → **disrupted NuRD-complex–dependent chromatin programs** controlling lymphocyte differentiation (inferred from AIOLOS molecular biology and shared Ikaros-family readouts).
4. → **Abnormal B- and T-cell development**: impaired humoral responses, high CD21low B cells, negative CD23, abrogated CD40 responses, decreased T cells, reduced recent thymic emigrants (demonstrated in patients).
5. → **Combined immunodeficiency** → recurrent sinopulmonary, bacterial, viral, and opportunistic infections (*Pneumocystis*; VDRV) (demonstrated).
6. **Parallel branch — immune dysregulation:** loss of AIOLOS restraint on lymphocyte activation (intestinal IELs; Th1 CXCR3 program; Th17 states) → **autoimmunity, allergy, SLE-like disease** (demonstrated in mice; inferred in patients).
7. **Late branch — malignancy:** chronic B-cell dysregulation/hyperproliferation → predisposition to **B-cell malignancy (CLL)** (observed in the index family; mechanism inferred).

Molecular pathways / processes. C2H2 zinc-finger transcription factor activity; NuRD chromatin remodeling; IFN-γ/STAT1 signaling (AIOLOS positively regulates CXCR3 on Th1 cells); lymphocyte differentiation and activation control. **Cell types involved:** B lymphocytes (CL:0000236), T lymphocytes incl. Th1 (CL:0000545), Th17 (CL:0002105), intestinal intraepithelial lymphocytes, naive CD4 T cells. **Suggested GO terms:** GO:0006355 (regulation of transcription, DNA-templated), GO:0030183 (B cell differentiation), GO:0030217 (T cell differentiation), GO:0002376 (immune system process), GO:0045580 (regulation of T cell differentiation). **Subcellular:** nucleus (GO:0005634), pericentromeric heterochromatin (GO:0005721).

7. Anatomical Structures Affected

- **Primary system:** hematopoietic/lymphoid and immune system (UBERON:0002405 immune system; UBERON:0002371 bone marrow; UBERON:0002106 spleen; UBERON:0002370 thymus).

- **Secondary involvement:** lungs (recurrent/opportunistic pneumonia; UBERON:0002048), gastrointestinal tract (diarrhea, IEL dysregulation; UBERON:0000160), kidney (immune-complex glomerulonephritis in mouse SLE model; UBERON:0002113), and CNS (VDRV with CSF involvement; UBERON:0001017).
- **Cell level:** B cells, T cells (Th1, Th17), intraepithelial lymphocytes, NK cells (by family analogy).
- **Lateralization:** systemic/bilateral — not a lateralized disorder.

8. Temporal Development

- **Onset:** typically **pediatric/childhood**, including infancy (G191R presented at 1 y 4 mo); some features (CLL) are adult-onset. Pattern is **chronic/insidious** with episodic infectious exacerbations.
- **Progression:** variable — from mild/incompletely penetrant to progressive and fatal (VDRV multi-organ failure). **Lifelong** disease.
- **Patterns:** no spontaneous remission; immune function can be stabilized with replacement/prophylaxis. **Critical period:** avoiding live vaccines in early childhood (before immune evaluation) is a key window to prevent catastrophic vaccine-derived infection.

9. Inheritance and Population

- **Inheritance:** autosomal dominant (heterozygous/monoallelic).
- **Penetrance:** variant-dependent — **full** for N160S; **incomplete** for E82K/Q402* haploinsufficiency.
- **Expressivity:** variable.
- **Origin:** familial segregation and de novo both documented; no anticipation, founder effect, or consanguinity dependence (dominant).
- **Epidemiology:** ultra-rare; **<~30 reported individuals** worldwide across a few families (as of 2024–2026). Precise prevalence/incidence undetermined. No strong sex bias established; both sexes affected. No defined geographic/ethnic clustering.

10. Diagnostics

- **Immunologic workup:** serum immunoglobulins (hypogammaglobulinemia; LOINC panels for IgG/IgA/IgM), lymphocyte subsets (decreased T cells, abnormal B-cell maturation: ↑CD21low, ↓CD23), vaccine/antibody responses, recent-thymic-emigrant quantification, CD40 response assays.

- **Genetic testing (definitive): whole-exome or whole-genome sequencing** / IEI gene panels including *IKZF3*; confirm with single-gene / segregation testing. CMA, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not indicated** (point variants).
- **Functional confirmation:** DNA-binding assays (EMSA), pericentromeric heterochromatin localization, dominant-negative/dimerization assays — these distinguish pathogenic variants and classify mechanism.
- **Differential diagnosis:** CVID; other Ikaros-family defects (IKZF1/Ikaros AD-CVID, IKZF2/Helios immune dysregulation); other combined immunodeficiencies and primary atopic disorders. Distinguish by gene and by the AIOLOS-specific functional signature.
- **Screening:** cascade genetic testing of at-risk relatives.

11. Outcome / Prognosis

Prognosis is **variable**. Many patients stabilize with immunoglobulin replacement and prophylaxis, but the disorder carries risks of **autoimmune morbidity, B-cell malignancy (CLL), and life-threatening/fatal infection** (e.g., disseminated VDRV with multi-organ failure and CNS involvement). No formal survival statistics exist for this ultra-rare disease. **Prognostic factors:** variant class/mechanism (DNA-binding dominant-negative appears more fully penetrant/severe), degree of T/B-cell compromise, and infection/malignancy history. Live-vaccine exposure is a major adverse prognostic event.

12. Treatment

- **Supportive/replacement: immunoglobulin replacement (IVIG/SCIG)** for hypogammaglobulinemia (NCIT: Intravenous Immunoglobulin Therapy); **antimicrobial prophylaxis**, including **trimethoprim-sulfamethoxazole** anti-*Pneumocystis* prophylaxis (NCIT: Trimethoprim-Sulfamethoxazole; CHEBI:9332 sulfamethoxazole).
- **Immunomodulation** for autoimmune/dysregulatory manifestations (corticosteroids, targeted immunosuppression; individualized).
- **Definitive/curative (severe phenotypes): allogeneic hematopoietic stem cell transplantation (HSCT)** (NCIT: Allogeneic Hematopoietic Stem Cell Transplantation) — inferred from general combined-immunodeficiency practice; not yet reported as a disease-specific trial.
- **Mechanistically contraindicated: IMiDs/CELMoDs** (lenalidomide, pomalidomide, iberdomide, mezigdomide) degrade AIOLOS via cereblon — opposite to the loss-of-function pathology.

- **Experimental/targeted:** none disease-specific; no NCT trials identified for this disease.

13. Prevention

- **Primary:** genetic counseling for at-risk families (50% transmission; variable penetrance; de novo possible); avoidance of live attenuated vaccines once immunodeficiency is suspected/confirmed.
- **Secondary:** early immunologic evaluation and genetic diagnosis; cascade/carrier testing; prenatal/preimplantation options via counseling.
- **Tertiary:** infection prophylaxis, immunoglobulin replacement, autoimmune management, and **malignancy surveillance** (CLL).
- **Immunization note:** inactivated vaccines generally acceptable; **live vaccines contraindicated** (VDRV risk).

14. Other Species / Natural Disease

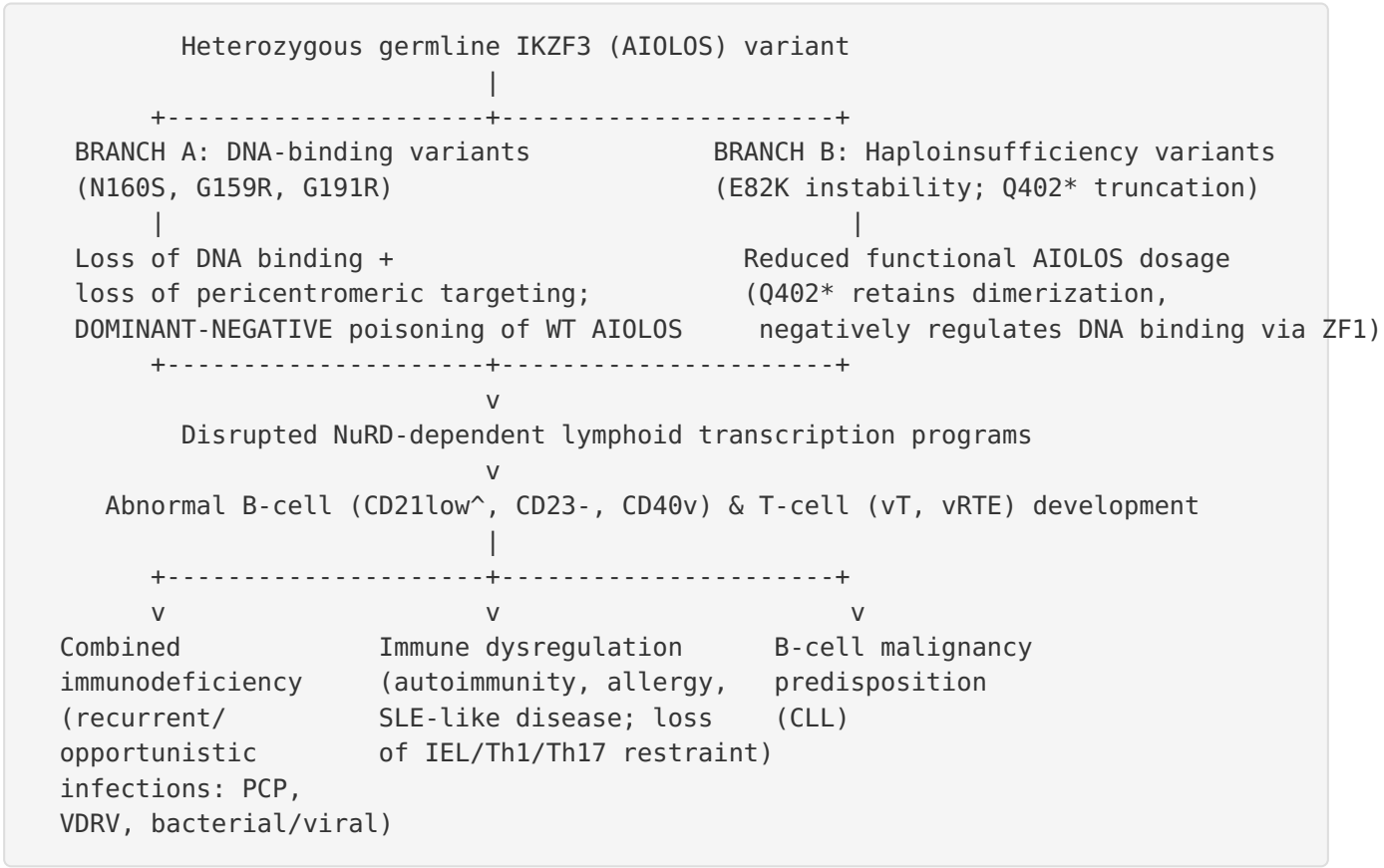
- **Orthologous gene:** mouse *Ikzf3* (Aiolos), NCBI Gene 22780. **Model species:** *Mus musculus* (NCBI Taxon 10090).
- **Natural disease in other species:** no well-characterized naturally occurring companion-animal/wildlife disease documented; relevance is primarily experimental. No zoonotic potential.
- **Comparative biology:** AIOLOS function and the SLE-like autoimmune consequence of its loss are conserved in mouse, supporting evolutionary conservation of the lymphoid transcriptional program.

15. Model Organisms

- **Aiolos-null mouse:** develops SLE-like autoimmunity (anti-dsDNA, immune-complex glomerulonephritis, B-cell hyperproliferation, spontaneous germinal centers) — models the **immune-dysregulation/autoimmune** arm (P 12574333).
- **Humanized knock-in mouse (N160S-equivalent):** recapitulates the **patient immunophenotype** — models the **dominant-negative immunodeficiency** arm (P 34694366).

- **Influenza-infection mouse model:** used to show Aiolos positively regulates CXCR3 on Th1 cells
(39560988).
- **Applications:** mechanism of lymphocyte differentiation, autoimmunity, and infection susceptibility. **Limitations:** mouse CRBN resists IMiD-induced AIOLOS degradation (relevant to pharmacology), and incomplete penetrance in human haploinsufficiency may not be fully captured. **Resources:** MGI (Ikzf3).

Mechanistic Model / Interpretation



The unifying interpretation is that **AIOLOS is a dosage- and function-sensitive master regulator of lymphocyte development and activation restraint.** Whether function is lost by poisoning (dominant-negative) or by dosage reduction (haploinsufficiency), the cell cannot execute normal B/T differentiation programs, simultaneously producing **immunodeficiency**

(failure of protective immunity) and **immune dysregulation** (failure to restrain autoreactive/inflammatory programs) — the hallmark "immunodeficiency + dysregulation" duality shared across the Ikaros family.

Evidence Base

PMID	Title (abbrev.)	Role	Evidence type
34694366	*T/B abnormalities, PCP, CLL with AIOLOS defect*	Defines disease; N160S dominant-negative; humanized mouse	Human + in vitro + mouse
38015619	*AIOLOS variants → haploinsufficiency; redefine domains*	Defines haploinsufficiency class (E82K, Q402*); 9 pts/3 families	Human + in vitro
38773004	*AIOLOS-Associated Inborn Errors of Immunity*	Review; variant catalog + infection phenotype	Review
38758229	*De novo IKZF3 (G191R) pediatric CID*	De novo origin; infantile combined ID	Human + in vitro
41746515	*Fatal VDRV granulomatous disease in AIOLOS ID*	Severe/fatal outcome; live-vaccine contraindication	Human
12574333	*Aiolos mutant mice → SLE-like disease*	Mouse autoimmune phenotype	Mouse
34920454	*IKZF2 (Helios) monoallelic mutations*	Places AIOLOS in Ikaros-family spectrum; shared dimerization/DNA-binding loss	Human + in vitro
38734037	*Novel IKZF1 variant, AD CVID*	Paralog IKZF1 haploinsufficiency; shared functional assays	Human + in vitro
39560988	*Aiolos promotes CXCR3 on Th1 via IFN-γ/STAT1*	Downstream effector program	Mouse + in vitro
38049581	*Aiolos restrains intestinal IEL activation*	Immune-dysregulation mechanism	Mouse
41922588	*Aiolos and Eos drive human TH17 states*	Human Th17 regulation by AIOLOS	Human in vitro
38539501	*Mezigdomide CELMoD*	Confirms IMiDs degrade AIOLOS (therapeutically opposite)	Review

How the evidence converges: The two foundational human papers (34694366, 38015619) establish the two mechanistic variant classes; the review (38773004) and additional case (38758229) extend the clinical/genetic spectrum and de novo occurrence; mouse models (12574333, 34694366, 39560988, 38049581) demonstrate causality and mechanism; and the paralog papers (34920454, 38734037) situate the disease within the Ikaros-family IEI landscape. No paper challenges the core causal model; the IMiD literature (38539501) clarifies that AIOLOS-degrading drugs are mechanistically inappropriate here.

Limitations and Knowledge Gaps

1. **Ultra-rare, small evidence base.** Fewer than ~30 reported individuals; genotype–phenotype correlations and penetrance estimates are based on a handful of families and may shift as more cases emerge.
2. **Missing formal identifiers.** OMIM, Orphanet, ICD, and MeSH IDs were not recorded/confirmed in this investigation; these should be cross-referenced.
3. **Epidemiology undefined.** No reliable prevalence/incidence, sex ratio, or geographic data.
4. **Treatment evidence is inferential.** No disease-specific trials; HSCT utility and immunomodulation choices are extrapolated from general combined-immunodeficiency practice.
5. **Mechanistic gaps.** The precise transcriptional targets linking AIOLOS loss to CLL predisposition, and the determinants of incomplete penetrance in haploinsufficiency variants, remain unresolved.
6. **Single-cell / omics data** specific to patient lymphocytes are limited; most mechanistic resolution derives from mouse and in vitro systems.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** for *IKZF3*-associated IEI to refine penetrance, natural history, infection/malignancy rates, and HSCT outcomes.
2. **Patient-derived single-cell multi-omics** (scRNA-seq + scATAC-seq) of B/T compartments to map AIOLOS-dependent regulatory programs and define biomarkers distinguishing dominant-negative vs. haploinsufficiency mechanisms.

3. **Systematic variant functional classification** (saturation DNA-binding/dimerization/pericentromeric-targeting assays) to triage future VUS and correlate mechanism with severity.
4. **Confirm/assign formal ontology identifiers** (OMIM, Orphanet, ICD-11, MeSH) and HPO frequency annotations for the knowledge-base entry.
5. **Evaluate HSCT timing and outcomes** in severe/fatal-risk patients via multicenter case collection; define malignancy-surveillance protocols for CLL.
6. **Reinforce live-vaccine contraindication guidance** in newborn/early-childhood pathways for families with known pathogenic *IKZF3* variants, given the fatal VDRV precedent.

Report compiled from 8 confirmed findings and 27 reviewed papers across 5 investigation iterations. Evidence types are labeled human clinical, model organism (mouse), in vitro, or inferred throughout.
